

Representing Sets and Operations Using Venn Diagrams

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A **Venn diagram** is a picture that shows the relationship between sets.

Sets in a Venn Diagram

A **set** is represented by a closed curve, while its **elements** are written inside the curve.

For example,

$$A = \{1, 2, 3\}$$

The elements 1, 2, and 3 are inside the curve representing A .

Subset

A **subset** is a set whose elements are all contained in another set.

If every element of B belongs to A , then

$$B \subseteq A.$$

For example,

$$A = \{1, 2, 3, 4\}$$

$$B = \{2, 4\}$$

Therefore,

$$B \subseteq A.$$

Union

The **union** of two sets, written as $A \cup B$, consists of all elements that belong to A , to B , or to both.

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

Therefore,

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

Intersection

The **intersection** of two sets, written as $A \cap B$, consists of all elements common to both sets.

$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

Therefore,

$$A \cap B = \{3\}.$$

Venn Diagrams

Venn diagrams help visualize the relationships among sets, subsets, unions, and intersections.

The overlapping region represents the elements common to both sets.

The combined regions represent the union.

Universal Set

The **universal set**, denoted by U , contains all elements under consideration.

For example,

$$U = \{1, 2, 3, 4, 5\}$$

If

$$A = \{2, 4\},$$

then A is a subset of U .

Examples

Example 1

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{3, 4, 5\}.$$

The elements common to both sets are 3.

Therefore,

$$A \cap B = \{3\}.$$

Example 2

Let

$$A = \{1, 2, 3\}$$

and

$$B = \{3, 4, 5\}.$$

All elements in either set are

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

Example 3

Let

$$A = \{2, 4, 6, 8\}$$

and

$$B = \{2, 4\}.$$

Every element of B is contained in A .

Therefore,

$$B \subseteq A.$$

Practice Exercises

Practice 1

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$,

$$A = \{2, 4, 6, 8, 10\}$$

and

$$B = \{1, 2, 3, 4, 5\}.$$

Draw a Venn diagram and shade $A \cup B$.

Practice 2

Using the sets in Exercise 1, draw a Venn diagram and shade

$$A \cap B.$$

Practice 3

Let

$$U = \{a, b, c, d, e, f, g, h\},$$

$$A = \{a, c, e, g\},$$

and

$$B = \{c, d, e, f\}.$$

Illustrate the sets using a Venn diagram and list the elements in each region.

Practice 4

Let

$$A = \{2, 4, 6, 8\}$$

and

$$B = \{2, 4\}.$$

Draw a Venn diagram to show that

$$B \subseteq A.$$

Practice 5

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\},$$

$$A = \{1, 2, 3, 4\},$$

and

$$B = \{5, 6, 7\}.$$

Illustrate the two disjoint sets using a Venn diagram.

Practice 6

In a class,

$$M = \{\text{Ana, Ben, Carl, Dana}\}$$

and

$$S = \{\text{Carl, Dana, Ella}\}.$$

Draw a Venn diagram and determine

$$M \cap S.$$

Practice 7

In a survey,

$$F = \{\text{apple, banana, mango}\}$$

and

$$G = \{\text{banana, mango, orange}\}.$$

Draw a Venn diagram and determine

$$F \cup G.$$

Practice 8

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\},$$

$$A = \{2, 4, 6, 8\},$$

and

$$B = \{4, 5, 6, 7\}.$$

Illustrate the Venn diagram and identify the elements belonging only to A .

Practice 9

Let

$$U = \{a, b, c, d, e, f, g\},$$

$$A = \{a, b, c, d\},$$

and

$$B = \{c, d, e, f\}.$$

Illustrate the Venn diagram and identify the elements belonging only to B .

Practice 10

Let

$$U = \{1, 2, \dots, 12\},$$

$$A = \{2, 4, 6, 8, 10, 12\},$$

and

$$B = \{3, 6, 9, 12\}.$$

Draw a Venn diagram and identify:

(a) $A \cap B$

(b) $A \cup B$

Activities

Activity 1

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$,

$$A = \{1, 3, 5, 7, 9\}$$

and

$$B = \{5, 6, 7, 8, 9\}.$$

Draw a Venn diagram and shade $A \cup B$.

Activity 2

Using the sets in Activity 1, draw a Venn diagram and shade

$$A \cap B.$$

Activity 3

Let

$$U = \{m, n, o, p, q, r, s, t\},$$

$$A = \{m, o, q, s\},$$

and

$$B = \{o, p, q, r\}.$$

Illustrate the sets using a Venn diagram and list the elements in each region.

Activity 4

Let

$$A = \{5, 10, 15, 20\}$$

and

$$B = \{10, 20\}.$$

Draw a Venn diagram to show that

$$B \subseteq A.$$

Activity 5

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\},$$

$$A = \{1, 2, 3\},$$

and

$$B = \{6, 7, 8\}.$$

Illustrate the two disjoint sets using a Venn diagram.

Activity 6

In a class,

$$M = \{\text{Leo, Mia, Noah, Pia}\}$$

and

$$S = \{\text{Noah, Pia, Ryan}\}.$$

Draw a Venn diagram and determine

$$M \cap S.$$

Activity 7

In a survey,

$$F = \{\text{grapes, apple, orange}\}$$

and

$$G = \{\text{orange, melon, apple}\}.$$

Draw a Venn diagram and determine

$$F \cup G.$$

Activity 8

Let

$$U = \{10, 11, 12, 13, 14, 15, 16, 17, 18\},$$

$$A = \{10, 12, 14, 16\},$$

and

$$B = \{12, 13, 14, 15\}.$$

Illustrate the Venn diagram and identify the elements belonging only to A .

Activity 9

Let

$$U = \{h, i, j, k, l, m, n\},$$

$$A = \{h, i, j, k\},$$

and

$$B = \{j, k, l, m\}.$$

Illustrate the Venn diagram and identify the elements belonging only to B .

Activity 10

Let

$$U = \{1, 2, \dots, 15\},$$

$$A = \{5, 10, 15\},$$

and

$$B = \{3, 6, 9, 12, 15\}.$$

Draw a Venn diagram and identify:

(a) $A \cap B$

(b) $A \cup B$

Thank you!